



JAD FALAQ

Machine Learning & MLOps Engineer

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6-month final-year engineering internship starting January 2027

PROFILE

Machine Learning & MLOps Engineer, final-year AI engineering student at **ENSIAS**, with experience at **Renault Digital** and **AKWA Group**. Specialized in **Machine Learning**, **Computer Vision**, **MLOps** and **model deployment**. Hybrid **Machine Learning / Ops** profile, focused on **model industrialization**, **production reliability** and **ERP integration**.

Machine Learning – Deep Learning – Computer Vision – MLOps – Data Engineering

PROFESSIONAL EXPERIENCE

AI Engineer Intern – Quality Engineering

Renault Digital | Internship | Casablanca, Morocco

Jun. 2026 – Aug. 2026

- **AI Agents & Test Generation:** Designed and deployed to production a multi-agent LLM system (Requirement Analyzer Agent, Test Design Agent) turning requirements into Jira tickets and Gherkin scenarios, with human validation at each step (human-in-the-loop). POC on Copilot Studio and Power Platform, then full-stack rebuild (FastAPI, React/TypeScript). 100+ Jira tickets and 50 Gherkin scenarios generated, writing time cut from 10–20 min to 1–2 min per requirement, adopted by a 6-person QA team.
- **RAG & Document Ingestion:** Built a RAG on the Microsoft Copilot API (cosine similarity, top-5) fed by an ingestion pipeline of 400+ PDFs (Confluence, past user stories and Jira tickets): OCR of tables and images (unstructured.io), document-structure chunking and MMR retrieval on pgvector. DLP masking (Presidio, spaCy) before every LLM call.
- **Evaluation & Monitoring:** JSON generation validated by Pydantic schemas, then deterministic Gherkin rendering checked by the official Cucumber parser, eliminating LLM syntax errors. Similarity threshold calibration and quality dashboard, with 85% of generations accepted without edits.
- **QA Automation:** Generation of Smoke/regression test files on SharePoint and Teams reporting (Power Automate), deployed internally.

ML & Data Engineer – Data Team

AKWA Group | Internship | Casablanca, Morocco

Jun. 2025 – Sep. 2025

- **Data Engineering & ERP Integration:** Automated, with the Data team, the detection and ERP integration of missing transactions across 6M+ SQL records, gaining hands-on experience with large-scale data processing and real-world data quality issues.
- **Machine Learning & Anomaly Detection:** Built an end-to-end ML pipeline (preprocessing, feature engineering, modeling, evaluation) to detect and classify missing transactions.
- **Modeling & Evaluation:** Compared 5 algorithms (logistic regression, SVM, decision tree, Random Forest, KNN); the selected Random Forest reached 98% accuracy and 85% F1-score.

TECHNICAL SKILLS

- **Machine Learning:** supervised and unsupervised learning (regression, SVM, Random Forest, XGBoost, KNN, clustering), dimensionality reduction and manifold learning, online and offline learning, reinforcement learning
- **Deep Learning:** CNN, LSTM, Transformers (encoders, decoders), GNN, fine-tuning (AraBERT)
- **Libraries:** NumPy, Pandas, scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, OpenCV, Ultralytics (YOLOv8), MediaPipe, Matplotlib
- **Data & Modalities:** time series, image/signal/video processing, 3D point clouds, Computer Vision (detection, object tracking, pose estimation)
- **MLOps & CI/CD:** MLflow (tracking, Model Registry), DVC, Airflow, Docker, Git, GitHub Actions, pytest
- **Serving, Data & Cloud:** Python, FastAPI, Pydantic, REST APIs, SQL, PostgreSQL, Azure

EDUCATION

State Engineering Degree – Artificial Intelligence Engineering (2IA)

ENSIAS, Rabat | Sep. 2024 – Jun. 2027

- Machine & Deep Learning (neural networks, reinforcement learning), statistical and probabilistic learning
- Big Data and distributed systems, data engineering, time series, multimedia processing (text, image)

Preparatory Classes for the French Grandes Écoles (CPGE)

CPGE Mohammed V, Casablanca | Sep. 2022 – Jun. 2024

- Applied mathematics, engineering sciences and Python programming

ACADEMIC PROJECTS

- **Computer Vision & Deep Learning:** Detection of 7 dental pathology categories (YOLOv8, 1,808 X-rays, 86% precision) / Real-time detection of suspicious behavior in video surveillance (YOLO, DeepSORT, MediaPipe Pose, LSTM; $\approx$ 80% accuracy)
- **NLP & Fine-tuning:** Fine-tuned AraBERT to detect medical misinformation in Arabic tweets, with deduplication, weighted loss, early stopping and threshold tuning, reaching 87% accuracy and 0.87 macro F1
- **MLOps:** Data/model versioning, experiment tracking (DVC, MLflow), FastAPI serving under Docker and GitHub Actions CI/CD for the dental model
- **CI/CD & Containerization:** Containerized (Docker) a LangGraph agent, GitHub Actions CI/CD and LLM tracing (LangSmith, Langfuse)

SOFT SKILLS

- **Organization & Teamwork:** Logistics lead of the ENSIAS Forum Club (company–student meetings)
- **Autonomy & Problem Solving:** Delivered an AI solution to production under access and confidentiality constraints

LANGUAGES

French: Native – English: Bilingual – Spanish: B1